

Statistical Diagnostics & Volumetric Integration

Goodness-of-Fit, Parameter Significance & Residual Analysis for Reservoir Water Depth Profile

GOODNESS-OF-FIT & ANOVA METRICS

METRIC	VALUE	DESCRIPTION
Coefficient of Determination (R^2)	0.979149	Proportion of variance explained by the model
Total Sum of Squares (SST)	2034.0779	Total variance sum (baseline data dispersion)
Sum of Squared Errors (SSE)	42.4125	Unexplained residual variance sum
Residual Standard Error (s)	0.3864 m	Standard deviation of residuals (DOF = 284)

PARAMETER SIGNIFICANCE & ESTIMATES

PARAMETER NAME	SYMBOL	ESTIMATE (B_j)	STD. ERROR (SE)	T-STATISTIC	P-VALUE
c (Baseline level)	c	14.3775	0.0379	379.3080	0.0000+00
a (Amplitude/Span)	a	5.5139	0.0499	110.4075	0.0000+00
k (Growth Rate)	k	0.8555	0.0625	13.6946	0.0000+00
t0 (Inflection Time)	t_0	30.1577	0.0978	308.2435	0.0000+00

VOLUMETRIC INTEGRATION COMPARISON

INTEGRATION METHOD	TOTAL EXPOSURE VALUE	CONTEXT
Analytical Quad (Fitted Model)	1260.9239 m·h	Continuous integration of 4PL curve function via SciPy quad
Trapezoidal Rule (Raw Telemetry)	1260.9963 m·h	Discrete numerical integration over all 288 data points

READING OF THE RESIDUALS

REGION / PHASE	OBSERVED PATTERN & INTERPRETATION
Baseline & Dip (0–32 h)	Tightly matched baseline data followed by a sharp negative excursion down to −0.4 (phase lag).
Peak Surge (33–45 h)	Major positive anomaly peaking near +1.3, showing the symmetric 4PL model under-predicts surge height.
Trailing Tail (>50 h)	Persistent negative bias plateauing at −0.4 where the model asymptotic tail sits above telemetry.